Syllabus Edition
First teaching 2025
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Using Break-Even Analysis to Make Decisions

Exam code: 0450, 0986 & 0264, 0774

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Updated on 28 August 2025

Break-even analysis and decision-making

- Break-even calculations are a useful tool for a business to use in deciding **how much to produce** and calculating estimated levels of profit
- It is particularly useful for **communicating with stakeholders**, including investors or lenders
  - Knowing when the business will break-even or how much profit it is expected to make **may attract or deter shareholders** from investing in the business
- Break-even analysis provides **a basis for informed decision-making**
  - It helps the business to assess the costs and expected returns of new projects and expansion plans
    - By considering the break-even point, businesses can **assess the potential risks and rewards** associated with different decisions

Ways break-even is used in decision-making

Use	How break-even analysis helps
Assessing profit or loss	• Shows the minimum sales needed to cover all costs and avoid losses • This helps the business set realistic profit targets
Managing costs	• Helps identify fixed and variable costs clearly

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Pricing decisions	• Supports pricing by calculating the minimum price required to cover costs and achieve the desired profit
Financial planning	• Acts as a guide for setting realistic sales targets and planning future spending
Redrawing the graph with changes	• Demonstrates how changes in costs, prices, or sales volume affect the break-even point • This helps assess risks such as price increases or new competition
Performance monitoring	• Provides a benchmark to compare actual sales and costs • This helps the business measure progress and financial health

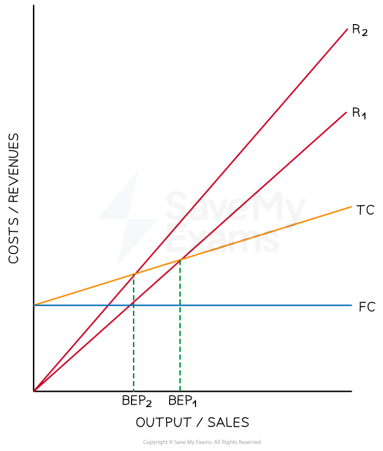
Help me understand

Changes to break-even variables

- Changing any of the variables of break-even (selling price, variable cost per unit or total fixed costs) changes **the break-even point and level of profit** it can expect to achieve

Increased selling price

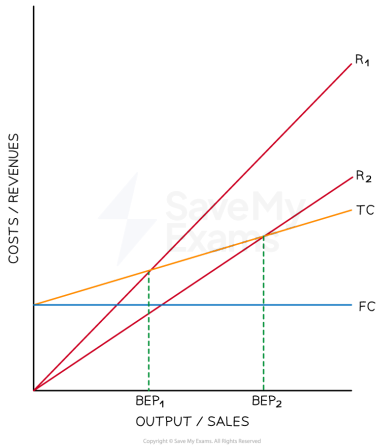
- An increase in the selling price **reduces** the break-even point



- An increase in the selling price increases revenue at each level of output from R_1 to R_2
- The break-even point falls from BEP_1 to BEP_2

Decreased selling price

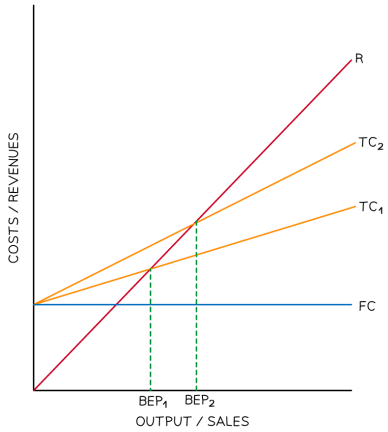
- A decrease in the selling price **increases** the break-even point



- A decrease in the selling price reduces revenue at each level of output from R_1 to R_2
- The break-even point rises from BEP_1 to BEP_2
- Profit on each unit of output greater than the break-even point is decreased

Increased variable costs

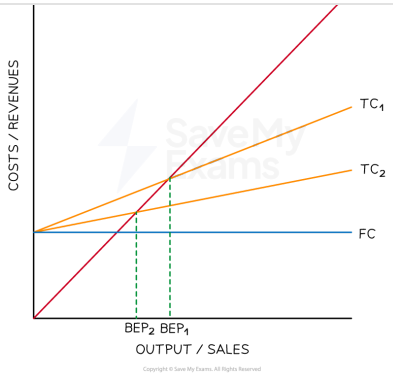
- An increase in variable costs **increases** the break-even point



- An increase in variable costs increases total costs at each level of output from TC_1 to TC_2
- The break-even point increases from BEP_1 to BEP_2
- Profit on each unit of output greater than the break-even point is decreased

Decreased variable costs

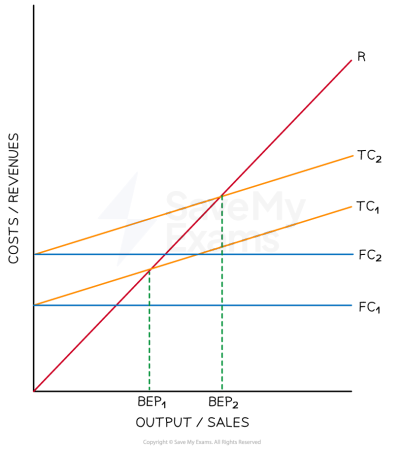
- A decrease in variable costs **decreases** the break-even point



- A decrease in variable costs decreases total costs at each level of output from TC_1 to TC_2
- The break-even point falls from BEP_1 to BEP_2
- Profit on each unit of output greater than the break-even point is increased

Increased fixed costs

- An increase in fixed costs **increases** the break-even point



- An increase in fixed costs increases total costs at each level of output from TC_1 to TC_2
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Decreased fixed costs

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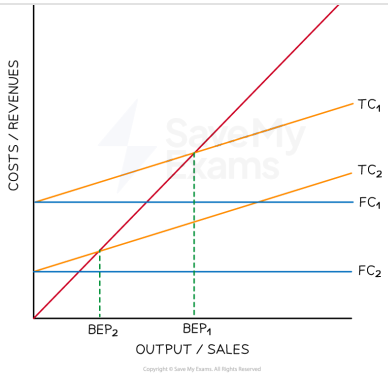

- A decrease in fixed costs reduces total costs at each level of output from TC_1 to TC_2
- The break-even point falls from BEP_1 to BEP_2
- Profit on each unit of output greater than the break-even point is increased

Examiner Tips and Tricks

Strengthen your answers by linking changes in **price, costs** or **sales volume** to shifts in the break-even point and margin of safety – this application gains higher marks

Limitations of break-even analysis

- Break-even analysis provides valuable insights into the **financial viability and performance of a business**
- The sooner a business can reach **break-even point**, the more likely it is to survive and make a profit
- However, there are several limitations to the use of break-even analysis

Limitations of break-even analysis include the assumption that all output is sold and its limited usefulness for multi-product businesses

1. Limited use for multi-product businesses

profit margins, it is difficult to calculate a single break-even point that applies to all of them

2. Relies on quality costs and revenue data

- The results are only accurate if the data used is correct
- If the business uses outdated or incorrect cost or revenue figures, the break-even point will be misleading

3. Assumes all output is sold

- Break-even analysis assumes that everything produced is sold with no leftover stock
- In reality, some products may not sell, which can make the analysis too optimistic

4. Revenue and costs are not always directly linked to output

- The model assumes that costs and revenue change in a straight line as output increases
- However, in real life, discounts, bulk purchasing or changing raw material costs can affect this relationship.

5. Requires skill to calculate and interpret correctly

- To get useful results, the business must understand how to calculate and read break-even charts and data
- If the analysis is done poorly, it can lead to wrong decisions

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